

RecognizingLicensePlates

Short project portfolio

Project goal

Build an educational automatic license plate recognition system from scratch in Python, showing every stage from image preprocessing to final OCR output without relying on OpenCV or scikit-image.

Technologies used

- Python with NumPy for all numeric image-processing and ML operations.
- Pillow for image file I/O and portfolio/demo image generation.
- Tkinter GUI for selecting images and inspecting recognition stages.
- Pytest for unit tests and sample benchmark verification.
- Gitflow-style workflow and pushed Git tags; GitHub Releases can be created separately.

Main features

- Desktop GUI for selecting an image directly from the computer.
- Step-by-step visualization: preprocessing, detection, normalization, segmentation, features, and OCR.
- Character segmentation into normalized 32x32 binary glyph crops.
- Blended OCR classifiers: synthetic MLP, pixel templates, zoning templates, and sample-template memory.
- No country-specific formatting is forced; the system focuses on raw character recognition.

Pipeline summary

1

Preprocess

grayscale, blur, CLAHE, Otsu

2

Detect plate

Sobel, morphology, connected components

3

Normalize

crop, Hough skew, affine rotation

4

Segment chars

cleanup, components, projection recovery

5

Classify

HOG, zoning, templates, MLP

Current verification

- Sample benchmark: 24/24 pass
- Unit tests: 95 passed
- Release archive: under 10 MB

GitHub link: <https://github.com/TqLinh30/RecognizingLicensePlates>

Portfolio generated from the local project repository.

Seven-Step Recognition Demo

Example image: data/samples/plate3.jpg. Final OCR result: 18A12345.

1 Input & preprocessing

The image is converted and contrast-enhanced.



2 Plate detection

Candidate boxes highlight likely plate regions.



3 Plate normalization

The plate is cropped, deskewed, and resized.



4 Character segmentation

Bounding boxes isolate individual characters.



5 Feature extraction

HOG + zoning descriptors are generated.



6 Classification

Model blending predicts each character.

OCR result

18A12345

7 Final output

Raw OCR text without country-specific formatting.

OCR result

18A12345

Repository

<https://github.com/TqLinh30/RecognizingLicensePlates>

Includes source code, tests, models, samples, docs, changelog, and Gitflow history.